

EE 371 Project Description

Project Description

In real FM broadcasting, the carrier typically lies around ~ 100 MHz. For the purposes of this project, however, a small portion of the FM broadcast band has been down-converted to a much lower and more manageable carrier frequency. This allows you to analyze and demodulate the signal using standard real-signal FM techniques without dealing with true RF-range frequencies, while still preserving the essential structure and behavior of a real FM transmission.

In this project, we will work with a *baseband FM signal* recorded using a software-defined radio (SDR). The aim is to analyze this signal in the frequency domain, identify its main components, extract the underlying audio information, and finally save the recovered audio as a sound file.

The dataset consists of a time-domain signal representing an FM broadcast channel. The sampling frequency of the provided signal is $f_s = 240$ kHz, and the target audio sampling rate is $f_{\text{audio}} = 48$ kHz.

- Plot the amplitude spectrum of the received baseband FM signal.
- Demodulate the FM signal to obtain the instantaneous frequency. Plot its amplitude spectrum and clearly identify the main FM message components.
- Design and apply a low-pass filter to isolate only the L+R (mono) component of the signal.
- Extract the message signal $m(t)$ corresponding to the L+R mono audio band, and plot its amplitude spectrum.
- Apply a de-emphasis filter to the mono signal and plot the resulting amplitude spectrum.
- Reconstruct the final audio signal, plot its amplitude spectrum, and save the output as a WAV file.

BONUS: Take a short segment from one of your favorite songs and FM-modulate it using the parameters $f_{\text{audio}} = 48$ kHz, $f_c = 110$ kHz, and $f_s = 240$ kHz. Save the modulated signal as `fav.mat`. If I can reconstruct the audio using these parameters and the sound is clear and intelligible, additional points will be awarded. (No stereo processing — simple FM modulation only.)

Note

All figures must be square, clearly labeled, include legends when necessary, and display correct physical units on all axes.

All figures must include descriptive captions. Captions should concisely explain what the figure shows and why it is relevant.

The maximum report length is 2 pages, including figures and references.